[[Lab Title]]: [[Accent Word]] [[rest of title]]

[[U0 / S0.3]]

NAME

DATE

PERIOD

GROUP MEMBERS

GOALS

- [[Operate a specific tool correctly — name the tool and the skill.]]
- [[Measure a specific quantity and explain why the wrong tool won't do.]]
- [[Determine / calculate the target quantity and state what you do with it.]]
- [[Build or set up the apparatus and accomplish the physical task.]]
- [[Record data in your lab notebook as you take it — not from memory afterward.]]

PURPOSE

[[You are going to ___ using nothing but ___. Both jobs depend entirely on whether you can use the tools correctly. That is the real point of today.]]

BACKGROUND

[[Concept 1 lead-in.]] [[Two to four sentences. Include the specific misstep students make — that sentence is what makes this section worth printing.]]

[[Concept 2 lead-in.]] [[Explanation.]]

> [[NAME OF THE RELATIONSHIP]]
>
> [[quantity = quantity ÷ quantity]] · [[units]]

[[One sentence on why the relationship behaves the way it does — the sentence that connects the math to the identification or conclusion.]]

> [[REACTION / EQUATION FOR TODAY]]
>
> [[$CaCl_2(aq) + Na_2CO_3(aq) \rightarrow CaCO_3(s) + 2\ NaCl(aq)$]]

> PERCENT ERROR
>
> percent error = (| experimental – accepted | ÷ accepted) × 100

SAFETY

- [[Splash goggles go on before any bottle is opened and stay on until every piece of glassware is washed and put away — including while other groups are still working.]]
- [[Chemical + concentration]] is [[hazard]]. On skin: [[exact response and duration]]. In eyes: [[eyewash duration, hold eyelids open, send a group member for me immediately]]. Do not wait to see if it gets better.
- [[Equipment-specific hazard and the action that prevents it.]]
- Check the rim of every [[piece of glassware]] before you use it. Chipped glass cuts. Broken glass goes in the glass box by the back sink — never the trash can.
- [[Thermal / electrical / burner statement, or: "Nothing is heated today and no burner is lit."]]
- Report every spill, including water. Water on the floor is how people fall.

PRE-LAB — DUE AT THE START OF THE PERIOD

Answer all of these in your lab notebook, numbered 1 through [[N]], before you walk in. Do not write on this handout — it stays clean so you can follow the procedure with wet hands. You will not be allowed to start the lab without the pre-lab finished, because most of these are the parts people get wrong when they wing it.

1. [[CONCEPTUAL — a distinction they will otherwise blur. Name the word you want used.]]
2. [[CONCEPTUAL — why one tool/method is trusted over another.]]
3. [[CONCEPTUAL — what has to be true for the observed phenomenon to happen, in their words.]]
4. [[PROCEDURAL — Read Part __. Exactly what quantity do you measure, and with what tool?]]
5. [[PROCEDURAL — In Part __, step __, where does ___ end up, and what problem does that solve?]]
6. [[PROCEDURAL — Which part do you start first, and why does that order matter for finishing on time?]]
7. [[PROCEDURAL — What do you do with ___ when you are finished with it?]]
8. [[SAFETY — Specific exposure scenario. State exactly what you do, and write where the eyewash station / extinguisher / shutoff is in this room.]]
9. Set up your [[N]] data tables in your notebook. Copy them exactly as shown, in ink, leaving enough room to write in each blank. Units go in the row labels — do not plan to add them later.

TABLE 1 · [[TABLE NAME]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]

TABLE 2 · [[TABLE NAME]]
[[Row label]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]
[[Row label with (units)]]

TABLE 3 · [[TABLE NAME]]
[[Qualitative observation row]]
[[Row label]]
[[Row label with (units)]]
[[next class]]
[[Row label with (units)]]

If you write a number wrong during the lab, draw one line through it and write the correct value beside it. Never scribble it out and never erase — I need to see what you originally recorded.

MATERIALS

Per group: [[item]] · [[item with size]] · [[item (2)]] · [[item]] · [[0.5 M reagent]] · [[0.5 M reagent]]

PROCEDURE

Part A — [[Part Name]]

1. [[Major step.]]
   - [[Substep — the check or condition.]]
   - [[Substep — what to do if the check fails.]]
2. [[Major step with the exact quantity and precision: "Record its mass to 0.01 g."]]
3. [[Major step.]]
4. [[Major step.]]
   - [[The thing they must write down before moving on.]]

Part B — [[Part Name]]

1. [[Major step.]]
2. [[Major step with reading precision: "Read at eye level and record to 0.1 mL."]]
3. [[The step that tells them not to "fix" their data: "Do not go back and adjust — the disagreement is the data."]]
4. [[Immediate cleanup step.]]

Part C — [[Part Name]] [[START THIS BEFORE PART D]]

1. [[Measure exact volume with the named tool and transfer to the named vessel.]]
2. [[Rinse / reset step between reagents.]]
3. [[The step where the thing happens.]]
   - Watch the instant [[they meet]]. Write what you see in your notebook before you do anything else — [[name the three properties: color, clarity, whether it settles]]. "It changed" is not an observation.
4. [[Build the apparatus:]]
   - [[Substep with a real dimension: "about 15 cm above the base."]]
   - [[Substep.]]
   - [[Substep + the reason it matters: "This stops ___ and pulls ___ through faster."]]
5. [[Major step.]]
6. [[Major step.]]
   - [[The failure mode stated as a consequence: "If it goes over the edge, ___ escapes into your ___ and your ___ is wrong."]]
7. [[Rinse-and-transfer step so nothing is left behind.]]
8. Go start Part D while this [[drains / runs / cools]].
9. [[What happens to the product at the end of the period and when it gets massed.]]

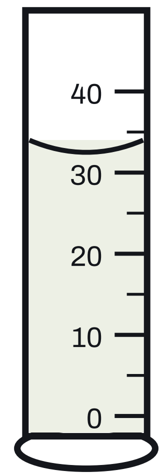
Part D — [[Part Name]] [[DO THIS WHILE PART C RUNS]]

1. [[Identify and describe the sample: color, luster, surface texture.]]
2. [[Measure to a stated precision.]]
3. [[Measure, with the anti-rounding warning: "Record the exact initial volume to 0.1 mL — not '30 mL.'"]]
4. [[Measure.]]
5. [[The subtraction / derived quantity.]]

6. [[Calculate. Carry units through the division.]]
7. [[Identify from the table below, then calculate percent error against the accepted value.]]
8. [[Return the sample to the tray.]]

[[MATERIAL]]	[[SYMBOL]]	[[ACCEPTED VALUE (UNITS)]]
[[name]]	[[sym]]	[[value]]
[[name]]	[[sym]]	[[value]]
[[name]]	[[sym]]	[[value]]
[[name]]	[[sym]]	[[value]]
[[name]]	[[sym]]	[[value]]

[[Conditions the values apply at, and any value that is a range rather than a single number — say why.]]



[[One-line reading instruction for the figure.]]

DATA TABLES

You built these in your pre-lab. Have them open and ready before you touch a [[balance / meter stick / hand lens]]. Fill them in **as you take each measurement** — not at the end of the period from memory.

DISPOSAL AND CLEANUP — LAST FIVE MINUTES

1. [[Liquid waste: exact destination and the reason it is permitted. Never imply — state it. "Our state permits acids and bases of 6.0 M and below down the drain, and this filtrate is dilute salt water — nowhere near that limit." VERIFY before printing.]]
2. [[Solid waste / product: where it goes, and when it is finally thrown out.]]
3. [[Reusable specimen or sample: how it is returned and in what condition.]]
4. **Wash all glassware with the Alconox solution in the blue spray bottle.** Spray inside, scrub with a brush, and rinse thoroughly — soap left in a beaker changes the next measurement made in it.
5. Rinse with distilled water and invert everything on the rack to dry.
6. Wipe your bench, push in your stool, and wash your hands. **Goggles come off last**, once the whole room is finished — not once you are.

POST-LAB ANALYSIS

Answer these in your lab notebook, numbered 1 through [[N]]. Show every calculation with units — an answer with no work and no units earns partial credit at best.

1. [[WHAT HAPPENED — two measurements disagreed. State which is closer to true and explain what about the two tools causes the difference.]] PART B
2. [[CALCULATION — show the calculation, name your result, and calculate percent error against the accepted value. Units in every step.]] PART D
3. [[WHY — using your recorded observations, explain where the product came from. Refer to the equation in the Background.]] PART C
4. [[ERROR ANALYSIS — a specific procedural slip. State whether it raises or lowers the calculated value, and say specifically whether the error entered through the ___ or the ___.]] ERROR ANALYSIS
5. [[ERROR ANALYSIS — why a step had to happen in the order it did. What would you be unable to determine if you skipped it?]] ERROR ANALYSIS
6. [[CONNECTION — pick one measurement you made today and identify the correct equipment for it from Section [[0.1]], then name the one safety rule from Section [[0.2]] that applied while you took it.]] [[U0 / S0.1 · S0.2]]